

Light and Strong

Graphene can lift a 4-kg cat on its 1-metre-square of its hypothetical area while weighing just 0.77 milligrams

What is graphene?

Think graphene can be used only in touchscreens? Have a look at this <http://dgit.in/J9v4XW>

FLEX YOUR DISPLAY

By all estimates, 2012 will witness the first wave of devices with flexible touchscreens. We take a close look at the technology

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It's been a while since Nokia's concept phone Morph teased our imagination on how the future of flexible displays and touchscreens would look like. Seemingly getting closer to their aim of flexing displays, they demonstrated their Kinetic Device at Nokia World 11 (<http://dgit.in/KamCEt>). It used flex-to-operate technology to browse through photos and music. Many other manufacturers, such as Samsung too showcased their innovations at CES 2011 wherein a curved S-shaped structure lit like an AMOLED display. Not only that, one could literally bend and fiddle with these displays that were as thin as human hair. But the real question is, are these display technologies going to replace the current ones anytime soon or are they only good enough for expos and sci-fi movies? Let's try to find out.

The story so far...

Most breakthroughs in technology have their roots in the development of material science. In the case of flexible touchscreens, the credit goes to a carbon allotrope known as graphene. So far, indium tin oxide (ITO) has firmly established itself as the most important ingredient in the touch screen market. This hard and brittle substance is scarce and becoming increasingly expensive. In the past six years, the price of this compound has shot up from \$100/kg to around \$1000/kg. The advent of graphene and the continuous development of the methods for its mass production might well change the way our next touchscreen phone fits in our pocket. Graphene is a single-atom thin sheet of carbon, which makes anything made out of it light and flexible. It's also



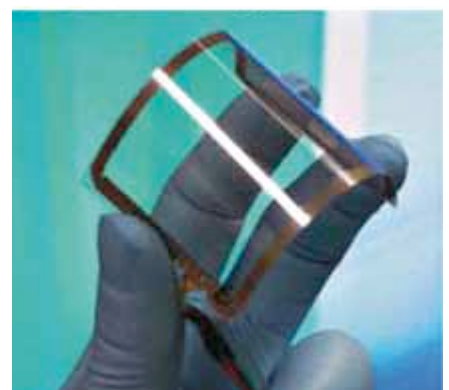
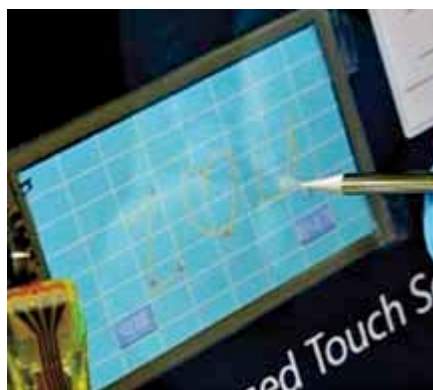
Atmel's XSense on an edgeless tablet

strong and highly conductive – qualities a researcher would ideally want. Surely there have been hurdles in its mass production since its discovery 40 years ago. Initially, graphene was too conductive to be used as an electrode. Attempts on producing large sheets of graphene, resulted in a change in its flexible and conductive properties. This in turn adversely affected the display quality when used to create touch screen. It created dead pixels which are a bane in display technology. Also, it was difficult to churn out single-

layers of large graphene sheets without damaging it. All these factors hindered the mass production of graphene so far. Clearly, its time hadn't come.

Why now?

The mainstream research for mass production of graphene gained momentum after 2004 when nobel laureates Andre Geim and Kostya Novoselov from the University of Manchester, UK, first suggested using adhesive tape to peel thin layers of graphene from lumps of graphite



Graphene sheet with silver electrode printed on it